

Lab 3 – Individual Report

Role and Contributions in Part 2

My main contribution to Part 2 was developing and testing gameplay mechanics for my 3D Star Defender 4-inspired game using Unity and C#. I worked on the player movement system, ensuring that the player could move horizontally along the bottom of the screen without moving outside the game boundaries. I also implemented the player shooting system and bullet collision detection. Invaders require two bullets to be destroyed and destroying them increases the player's score.

I contributed to the enemy systems by implementing three different waves. The first wave consists of invaders moving together as a formation and shooting downward. The second wave features a circular movement pattern, while the third wave contains a more dangerous invader that moves freely and fires projectiles at different angles. I also worked on wave progression so that the next wave starts when the previous wave is completed.

Another contribution was implementing the player's health system, including three lives, respawning, and flashing during a two-second invulnerability period. I also developed a power-up system that temporarily upgrades the player's weapon to fire three bullets simultaneously for ten seconds. Additionally, I worked on the score and lives displays and the victory message shown after completing all three waves.

Problems Faced and Solutions

One challenge was making the invader formation respawn correctly after all its enemies were destroyed. Initially, the formation remained in the Hierarchy without any children, preventing the wave system from recognizing that the wave had finished. I solved this by updating the wave-spawning logic to detect when the formation had no remaining invaders.

Another problem was progressing from the first wave to the second wave. The formation resawner interfered with the wave manager, so I disabled the separate resawner to allow the waves to progress in the correct order. I also tested the score, lives, and power-up systems to ensure they behaved as expected.

Collaboration and Evaluation

Working on this project helped me understand how different gameplay systems can work together in a 3D game.

Overall, I improved my understanding of C# scripting, collision detection, object spawning, coroutines, UI updates, and game-state management in Unity. The project also taught me the importance of testing individual features before integrating them. Further improvements include completing the required audio effects.